

## Natural Sounds

**Grades:** Primary, Middle

**Objectives:** As a result of completing this activity, students will:

(Art) Use music and language to communicate auditory observations.

(Science) Understand the importance of sound in observing the natural world, and birds in particular.

**Materials:**

- audio tape with bird songs and calls
- tape player
- drawing materials—paper, colored pencils, crayons, etc.
- Examples of poetry which feature birds

**Instructions:**

**Activity I: Bird Songs**

1) Hold a general discussion of the role of sound in our daily lives and in nature. Ask students which sense they rely on most heavily—smell, hearing, touch, or sight? Explain that bird watchers develop a strong sense of sight to identify birds by discrete markings such as color of wing bars and eye rings. Bird watchers also develop a strong sense of hearing to identify birds by their songs or calls. In fact, many expert bird watchers rely on their hearing to a greater degree than sight to identify birds.

2) Ask students to think about the bird songs and calls they have heard.

3) Darken the room. Tell students you will play a tape of bird songs. Encourage students to try to identify as many songs or calls as possible. After five minutes of listening, pause the tape and ask students the following questions:

- a) How many different birds did they hear?
- b) What other sounds did they hear in the background?
- c) Which bird songs had they heard before?
- d) Can they identify any of the birds on the tape?

**Activity II: Onomatopoeia Poetry**

1) Onomatopoeia (on-ah-mat-ah-PEE-ah) means to use language to describe what we hear. Words like *swish*, *thump*, and *gurgle* are onomatopoeias. Ornithologists use this term to describe bird calls or songs. For example, the song of the American robin sounds like “cheerily, cheerrup, cheerio, cheerily!” Even the



apparent distance away.

3) Allow approximately 10 minutes for listening and recording.

Back in the classroom, hold a discussion to identify the loudest, the softest, most pleasant, mysterious, or annoying bird songs.

4) Pass out colored pencils, crayons, and other drawing materials. Have students complete their sound maps by "coloring in" the spaces between the symbols with other elements of the habitat. Display sound maps in the classroom.

#### **Activity IV: Bird Music**

1) Tell students you will play the tape one more time and have them listen very closely to one song.

2) Have students who play an instrument such as the flute, obce, clarinet, trumpet, etc., try to emulate the bird's song on their instrument. Even if they do not own and play an instrument, students can produce the song's "beat" with their hands or simple percussion instruments. You may wish to work with a music teacher on this exercise. With some help, students might even try to develop simple musical notation of the bird's song.

#### **Activity V: A Symphony of Animal Sounds**

1) Obtain a tape of "Peter and the Wolf" by the Russian composer Serge Prokofiev (1891-1953) which tells the story of the boy, Peter, and his three friends—the bird, the duck, and the cat.

2) Explain that "Peter and the Wolf" includes various musical instruments to represent animals: the bird, by a flute; the duck, by an oboe; the cat, by a clarinet; and the wolf, by three french horns. Ask students why they think Prokofiev selected the instruments he did for each character.

3) Play a portion of the recording and have students decide whether they want to perform the part of Peter, the wolf, the bird, the duck, or the cat.

4) Resume the recording and allow students to perform the role of Peter, the wolf, the bird, the duck, or the cat. Play recordings of other instruments and ask the students to describe what animals might be represented with each instrument.



names of some birds are onomatopoeias, such as veery, jay, killdeer and chickadee.

Ask students if they can think of any birds named by onomatopoeia. Note the examples included in the Background section and ask students to list any others they can think of. Tell them that poets use onomatopoeia to enliven their poetry. An example by the famous naturalist John Burroughs follows in which he refers to the bluebirds "turalee" flight note:

### Bluebird

A wistful note from out the sky,  
"Pure, pure, pure," in plaintive tone  
As if the wand'rer were alone,  
And hardly knew to sing or cry.

2) Play the tape again. Ask students to pay attention to any onomatopoeias they hear (for example—*swish*, *thump*, and *gurgle*). Tell them to listen carefully because they will be writing short poems using onomatopoeias.

3) Assign students' poems using onomatopoeias either as an in-class writing exercise or as a take home writing assignment. You could also have students write their onomatopoeia poems in small cooperative working groups. If you work with younger students, you could develop your onomatopoeia poems as a class.

### Activity III: Sound Maps

1) Distribute 8 1/2 X 11" paper and have students draw a large circle on a blank sheet, placing an "x" in the center to indicate their location as shown on the diagram.

2) Take students back to the location of your sensory field trip, (or to another predetermined location).

Relative to their position, have them diagram the location of each sound they hear inside the circle.

They will use simple representative sketches or symbols to indicate the bird songs and other sounds as indicated on the diagram. Have them try to note the direction the song or sound is coming from, its loudness, pitch, melody, and

